R20

Code No.: 20CSC22

CHAITANYA BHARATHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.E. (CSE, CSM) V Sem (Main & Backlog) Examination Dec 2023 – Dec 2023

Software Engineering

Time: 3 Hours **Max Marks: 60**

Note: Answer **ALL** questions from **Part-A & Part –B (Internal Choice)** at one place in the same order

Part - A
(5Q X 3M = 15 Marks)

			M	CO	BT
1		List the different approaches for software process assessment.	(3)	1	1
2		What is an Agile Process?	(3)	2	1
3		List the Problem based estimation guidelines.	(3)	3	4
4		Define the Open-Closed principle with a diagram.	(3)	4	1
5		What is System testing? Identify an example.	(3)	5	3

Part – B
(5Q X 9M = 45 Marks)

			M	CO	BT
6	(a)	Explain Waterfall model with a diagram.	(5)	1	2
	(b)	Demonstrate Software Engineering layers with a neat diagram.	(4)	1	2
		(OR)			
7	(a)	Explain Software Process Framework.	(5)	1	2
	(b)	Construct the Spiral model with a diagram.	(4)	1	3
8	(a)	Illustrate Adaptive Software Development in Agile Process Model.	(5)	2	2
	(b)	List and explain the Collaborative Requirements Gathering guidelines.	(4)	2	4
		(OR)			
9	(a)	Identify and explain the Non-Functional requirements of an SRS.	(5)	2	3
	(b)	Illustrate the Validating requirements.	(4)	2	2
10	(a)	Distinguish LOC-Based and FP-Based Estimation.	(5)	3	4
	(b)	What is flow-oriented modeling in the context of software development?	(4)	3	1
		(OR)			
11	(a)	What are the different steps involved in risk management?	(5)	3	1
	(b)	Explain different strategies for staffing a software project?	(4)	3	2
12	(a)	Identify the different levels of cohesion and their implications for code comprehension and reusability.	(5)	4	3
	(b)	Illustrate Component –level Design guidelines.	(4)	4	2
		(OR)			
13	(a)	List and explain briefly Taxonomy of Architectural Styles.	(5)	4	4
	(b)	Examine the importance of using appropriate data structures and algorithms when designing software components.	(4)	4	4

14	(a)	Explain White Box Testing Methods.	(5)	5	2
	(b)	Distinguish between informal reviews and formal technical reviews.	(4)	5	4
		(OR)			
15	(a)	Define the term "software quality" and explain its significance in software development and user satisfaction.	(5)	5	2
	(b)	Identify and explain the importance of test case design in software testing and the factors to consider when designing effective test cases.	(4)	5	3

******